

A bounded-founder alternative that allows late founders

A second CA-free answer to the coverage question of Alexander's *Specieslike clusters* (2026), and a correction to note 06

27 September 2026

Provenance. Produced by OpenAI Codex during an external audit of this repository (27 September 2026), and checked by Anthropic Claude before inclusion. Hand proofs; no independent human review; priority not established.

The conclusion after Proposition 4 in note 06 is too strong: the immigrant example does **not** imply that every CA-free coverage solution must exclude every late founder. It implies that an IAP-contained class cannot admit all the successive immigrant extensions in that example and still have the desired maximality property.

Here is a different constraint that permits late founders while retaining upward genericity and coverage.

Definitions

Use Alexander's infinite-biosphere assumptions: parents precede children, finitely many vertices precede every real time, each vertex has finitely many children, and the graph is infinite. A founder of S has no parent in S . Work with nonempty clusters throughout. For a fixed integer $N \geq 1$ put

$$\mathcal{F}_N = \text{CONN} \cap \text{IAP} \cap \text{CONV} \cap \text{REF} \cap \{S : S \text{ has at most } N \text{ founders}\}.$$

CONN is undirected connectedness; IAP, CONV, REF have the definitions in note 10. The bound N is fixed for the whole class, not chosen anew for each member.

Theorem

For every $N \geq 1$, $\mathcal{F}_N$ is closed under nonempty increasing-chain unions, and every organism lies in an inclusion-maximal member. For $N=1$ it is exactly Alexander's $\text{IAP} \cap \text{CONV} \cap \text{CA} \cap \text{REF}$ class. For $N \geq 2$ it admits maximal clusters with no common ancestor and with a founder born after a non-founder.

Proof of closure. Let C be the union of an increasing chain C_α in $\mathcal{F}_N$. CONN, CONV and REF pass to C by the usual finite-witness and inclusion arguments. If C had $N+1$ founders, some stage would contain all of them. They would remain founders there, contradicting its bound N .

Every vertex of C descends from an internal founder or is one: internal parent chains terminate by finite ancestry. Suppose IAP fails at $v \in C$. There are infinitely many descendants and

non-descendants of v in C . Among the at most N founders, some w_1 has infinitely many descendants in C that do not descend from v . Finite branching and convexity produce an infinite path $w_1, w_2, \dots$ in C , none descending from v .

Choose α containing v, w_1 . REF makes C_α infinite. For each $n > 1$ choose $\beta \geq \alpha$ containing w_n . REF gives w_n infinitely many descendants in C_β , so IAP makes its non-descendants in C_β finite. The infinite subset C_α must contain a descendant of w_n . It already contains its ancestor w_1 ; convexity therefore puts w_n in C_α . Thus C_α contains infinitely many non-descendants of v . REF also gives it infinitely many descendants of v , contradicting its IAP. Closure follows.

Coverage. Alexander's Theorem 13 supplies each organism with a containing CA/CONV/IAP/REF cluster. CA and convexity give exactly one founder and connectedness, so that cluster belongs to $\mathcal{F}_N$. Chain closure and Zorn's lemma extend it to a maximal member.

N=1. A single founder is the ancestor of every other included organism, by termination of internal parent chains. Conversely CA and convexity force the common ancestor to be the unique founder. This proves the equality of classes. ■

Explicit late-founder example

Take vertices $a_0, a_1, a_2, b, c_0, c_1, \dots$ in that birth order, with edges

$$a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow c_0 \rightarrow c_1 \rightarrow \dots, \quad b \rightarrow c_0.$$

Let birthdates be consecutive integers and let S be the whole biosphere. Its founders are a_0 and b . Founder b is later than the non-founders a_1, a_2 , so **FC fails**. Yet every vertex has all but finitely many vertices as descendants: IAP holds. Connectedness, convexity and reflection hold for $S=G$. It belongs to $\mathcal{F}_2$, and is maximal because it is all of G . It has no common ancestor: neither founder descends from the other.

Thus fixed finite founder budgets, unlike the FC time condition, can admit late immigration. The choice of N is an extra convention whose biological merit is not established by the proof.

Allowing "some finite number of founders" **without one bound N for the class** does not give this argument: a chain can accumulate infinitely many founders. The paper's immigrant counterexample shows why that difference matters.

Infinite species classes still agree

The proof of Theorem 1 of note 10 uses finiteness of founders, REF, IAP and CONV; it does not use the FC birth-time inequality after finiteness is established. Consequently, for each $N \geq 1$, the infinite maximal members of $\mathcal{F}_N$ represent exactly the same generator-equivalence classes as the infinite CA-maximal members.

This is another answer to the CA-free coverage request after Theorem 13 of [Alexander's paper](#). It changes which finite founding histories are admitted while preserving the represented infinite species classes. It does not identify finite clusters under the paper's infinite-set equivalence relation.